KREEDA

Sports management system

1. Problem. Courts at the PES sports complex are in far more demand than there is court time. Slots are claimed informally at the counter, so students argue over who booked first, the same slot can be promised twice, and a court often sits empty because whoever claimed it did not turn up while others were turned away. Nothing is recorded, so nobody can say which hours are busiest or who repeatedly books and does not show. Tournaments are announced on notice boards, with no list of who registered and no link to the courts they occupy, and damaged nets are reported verbally and forgotten.

2. Objectives. Move booking online, with the database guaranteeing that one court slot belongs to exactly one booking. Apply fair-use limits so nobody blocks peak hours daily, keep a waiting list that moves up when a booking is cancelled, and mark no-shows. Manage events in the same place, so a tournament blocks the courts it needs and clashing bookings are cancelled and notified. Log complaints, group repeated ones, and report court usage by hour and day.

3. What the system does. Members see live availability, book and cancel slots, join a waiting list, post what game they are looking for, register for events, and raise complaints. Staff mark attendance against bookings, issue and return keys and equipment, and act on complaints. Admin sets court timings and booking rules, creates events and blocks courts for them, and views usage, no-show and complaint reports.

4. Databases used. PostgreSQL for the main data, with the pgvector extension for the free-text parts.Also using Qdrant for vector db storage. This is Option 2 in the project requirement. Booking is why the relational side matters: two students can press book on the same slot in the same second, and only the database can decide the winner, using a transaction and a rule that refuses any booking overlapping an existing one. Three parts of the system are free text that people word differently every time, where keyword search fails and meaning-based search is needed.

5. Database features we will implement

Feature	**Used for**
Transactions	Booking, cancelling, waiting-list promotion, blocking courts for events
Constraints	No two bookings on the same court and time; booking limits; event capacity
Triggers	Freeing cancelled slots, marking no-shows, recording who changed what
Stored procedures	Booking, cancellation and event rules, kept inside the database
Views	Live availability, today's schedule, open complaints
Indexes	Fast lookups on court, date and member
Vector search	Find a game: "casual doubles, in evening time" matched to " games happening after 6pm”
	Repeat complaints: "net torn on court 2" matched to "badminton net problem, court two”
	Event suggestions: handling the timings during ongoing tournaments , blocking etc

7. Database design(starting version): An ER diagram covering members, staff, courts, slots, bookings, waiting lists, attendance, events, registrations, complaints, keys and equipment, converted into tables in third normal form with keys and constraints. The diagram and table list come with the report.

8. Web application and security to an extent: Next.js for the interface and the API layer that talks to the database. Each role sees only what it is allowed to. Passwords are stored encrypted, and database credentials sit in environment variables instead of the code.

9. Work split for now: Dhanush Reddy: database design, tables, booking and event rules, triggers, stored procedures, indexes and the search features. Dhanush SS: Next.js pages, login and roles, booking screens, dashboards, reports, test cases and documentation. We will commit to a shared GitHub repository.

10. Why this is not a simple CRUD project?? The hard part is not storing a booking, it is deciding who gets the slot when two and keeping waiting lists, cancellations, no-shows, limits and event blocks consistent while that happens. Those rules live in the database, not in the forms, and the reports and search features are built on the same data.

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